



Institutional NAC Pre-Vetting & Risk Mitigation Framework

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Project: PoC 21 - Portfolio Optimization & Risk Mitigation

Tech Stack: Python | Streamlit | LangChain (RAG) | ChromaDB | Groq (Llama 3.3-70B)



Project Vision

In institutional banking, the New Activity (NAC) process is often a strategic bottleneck. This Proof of Concept (PoC) demonstrates how Retrieval-Augmented Generation (RAG) can shift governance from a reactive "No" to a proactive "Pre-Audit."

The tool acts as a Digital First-Responder, allowing Managing Directors to vet partnership proposals against complex internal policies in seconds, identifying regulatory "red flags" and financial misalignments before they reach the committee.



Executive Summary

Strategic Intent: To transform the 'New Activity' (NAC) process from a reactive bottleneck into a proactive strategic filter.

The Problem

Institutional deal teams often spend weeks preparing partnership proposals, only to face "late-stage rejection" due to governance or risk misalignments. This lag increases operational friction and slows the institutional deal pipeline.

The Solution

To eliminate the bottle neck, this PoC proposes the use of a Retrieval-Augmented Generation (RAG) framework that acts as a "Digital First-Responder." By cross-referencing multi-million-dollar deal proposals against internal bank policies in real-time, the tool identifies regulatory 'red flags' and provides an instant go/no-go risk assessment of a partnership proposal.

Key Value Drivers

- **Instant Visibility:** A dynamic Risk Heatmap provides immediate clarity on Financial, Regulatory, and Operational exposure.
- **Explainable AI (XAI):** Each finding is backed by an indisputable audit trail, quoting exact policy snippets from the internal knowledge base.
- **Operational Efficiency:** Drastically reduces committee review cycles by ensuring only "governance-ready" proposals move forward.

✨ Key Features

- **Dual-Engine RAG:** Combines deep narrative audit analysis with a dedicated quantitative scoring engine.
- **Risk Exposure Heatmap:** A color-coded visual dashboard (Yellow-Orange-Red) for instant risk quantification.
- **Indisputable Audit Trail:** Every finding includes Source Snippets—direct quotes from internal governance PDFs to ensure 100% transparency.
- **Executive Export:** Generates professional, board-ready PDF audit reports for formal submission.
- **Scalable Knowledge Base:** Modular design allows for instant policy updates via simple PDF ingestion without model retraining.

🔧 Technical Architecture

- **Orchestration:** LangChain (RAG Architecture)
- **LLM:** Llama 3.3-70B (via Groq) for high-speed institutional reasoning.
- **Vector Database:** ChromaDB (Local persistent storage).
- **Embeddings:** HuggingFace all-MiniLM-L6-v2 (100% local/private processing).
- **Frontend:** Streamlit (Optimized for Managing Director-level UX).



Scalability Roadmap: From PoC to Production

- **Multi-Agent Review:** Deploy specialized "AI Agents" for specific risk domains (e.g., a "Legal Agent" and a "Financial Audit Agent") to provide deeper cross-functional analysis.
- **Enterprise Auth:** Integration with Okta/Azure AD for secure, role-based access to sensitive governance documents.
- **Audit Trail Logging:** Connecting the tool to an immutable database (e.g., SQL or Blockchain) to log every pre-audit for future regulatory reviews.
- **Feedback Loop:** Allowing Risk Officers to "thumbs up/down" AI findings to fine-tune the RAG retrieval precision over time.



Installation & Usage

1. Clone & Setup:

```
pip install -r requirements.txt
pip install fpdf2 seaborn matplotlib pandas
```

2. Launch Portal:

```
streamlit run app.py
```

Use code with caution.

- ### 3. Analyze:
- Enter your Groq API key, ingest your policy library (/data/governance_docs), and upload a proposal to generate your first audit.



Dashboard Strategy

The UI is optimized for a Managing Director's review, and uses a 1:2 layout ratio that prioritizes a narrow, high-impact Risk Dashboard on the left and a detailed, high-readability Policy Narrative on the right. The policy narrative can be downloaded into a .pdf file format.

User Guide: NAC Pre-Vetting Portal

Role: Self-Service Strategic Alignment & Risk Identification

1. Getting Started

- **Authentication:** Enter your Groq API Key in the sidebar. This connects the secure "reasoning engine" to the portal. (A Groq API Key may be obtained by accessing the GroqCloud Console (<https://console.groq.com/home>) From this page a user can create a free account using their e-mail address, existing Google e-mail, or GitHub credentials.
- **Knowledge Sync:** If new governance policies have been released, click "Update/Ingest Governance Docs." The system will scan the internal library and refresh its "memory" (this takes about 30–60 seconds).

2. Running an Assessment

- **Upload:** Drag and drop your Partnership Proposal (PDF or TXT) into the upload zone.
- **Analyze:** Click  Run Risk Assessment."
- **What happens next:** The AI performs two tasks. First, it searches the policy library for relevant rules. Second, it generates the detailed audit and calculates the risk scores.

3. Reading the Dashboard (Left Column)

- **The Risk Bar:** Look at the heatmap for an immediate "Go/No-Go" signal.
 -  Green (0-3): Strong alignment. Likely to pass committee with minimal questions.
 -  Yellow (4-6): Potential friction. Requires the deal team to provide additional justification or waivers.
 -  Red (7-10): Policy breach. The proposal likely violates a "Hard Rule" (e.g., Data Sovereignty or IRR floors) and should be revised before submission.
- **Quick-Check:** Use the icon list for a 5-second summary of the three core risk categories.

4. Reviewing the Deep-Dive (Right Column)

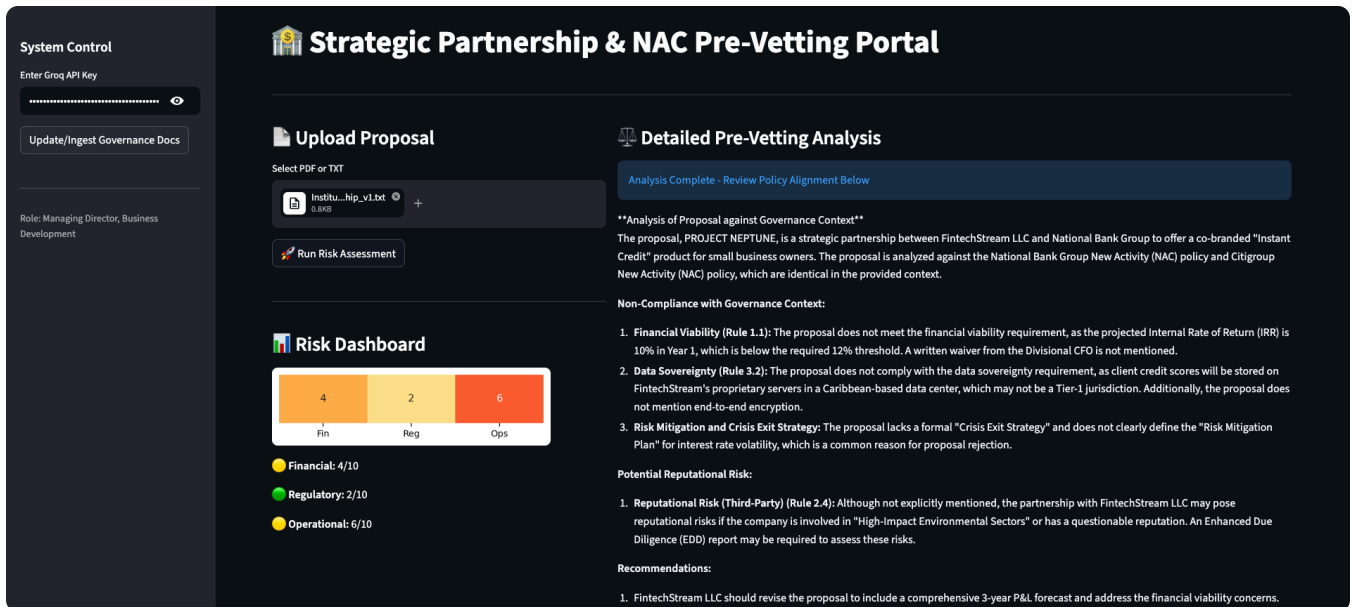
- **Executive Summary:** Read the top section for the AI's final verdict and a high-level recommendation.
- **Policy Snippets:** This is the most critical section. Look for the "Source Snippets" where the AI quotes the exact bank rule it is using to flag your proposal. This ensures the feedback is objective, not subjective.
- **Actionable Feedback:** Use the bullet points at the bottom of the report to refine your proposal. These are the specific steps needed to turn a "Red" or "Yellow" score into a "Green."

5. Exporting for Formal Submission

- **Download PDF:** Once you are satisfied with the alignment, click " Download Audit Report."
- **Best Practice:** Attach this PDF to your formal NAC submission. It demonstrates to the Risk Committee that you have performed pre-submission due diligence and have already addressed potential misalignments.



Screenshots



The screenshot displays the 'Strategic Partnership & NAC Pre-Vetting Portal' interface. On the left is a 'System Control' sidebar with fields for 'Enter Groq API Key' and 'Update/Ingest Governance Docs', along with a role assignment section. The main content area is divided into three sections: 'Upload Proposal' with a file selection box showing 'Institu...hip_v1.txt' and a 'Run Risk Assessment' button; 'Risk Dashboard' with a bar chart showing scores of 4 (Fin), 2 (Reg), and 6 (Ops), and a legend for Financial (4/10), Regulatory (2/10), and Operational (6/10); and 'Detailed Pre-Vetting Analysis' which includes a status bar 'Analysis Complete - Review Policy Alignment Below', a summary of the proposal analysis, a list of non-compliance items with specific rule references and descriptions, a section on potential reputational risk, and a final recommendations section.

System Control

Enter Groq API Key

Update/Ingest Governance Docs

Role: Managing Director, Business Development

Strategic Partnership & NAC Pre-Vetting Portal

Upload Proposal

Select PDF or TXT

Institu...hip_v1.txt (0.8KB)

Run Risk Assessment

Risk Dashboard

Category	Score
Fin	4
Reg	2
Ops	6

● Financial: 4/10
● Regulatory: 2/10
● Operational: 6/10

Detailed Pre-Vetting Analysis

Analysis Complete - Review Policy Alignment Below

****Analysis of Proposal against Governance Context****

The proposal, PROJECT NEPTUNE, is a strategic partnership between FintechStream LLC and National Bank Group to offer a co-branded "Instant Credit" product for small business owners. The proposal is analyzed against the National Bank Group New Activity (NAC) policy and Citigroup New Activity (NAC) policy, which are identical in the provided context.

Non-Compliance with Governance Context:

- Financial Viability (Rule 1.1):** The proposal does not meet the financial viability requirement, as the projected Internal Rate of Return (IRR) is 10% in Year 1, which is below the required 12% threshold. A written waiver from the Divisional CFO is not mentioned.
- Data Sovereignty (Rule 3.2):** The proposal does not comply with the data sovereignty requirement, as client credit scores will be stored on FintechStream's proprietary servers in a Caribbean-based data center, which may not be a Tier-1 jurisdiction. Additionally, the proposal does not mention end-to-end encryption.
- Risk Mitigation and Crisis Exit Strategy:** The proposal lacks a formal "Crisis Exit Strategy" and does not clearly define the "Risk Mitigation Plan" for interest rate volatility, which is a common reason for proposal rejection.

Potential Reputational Risk:

- Reputational Risk (Third-Party) (Rule 2.4):** Although not explicitly mentioned, the partnership with FintechStream LLC may pose reputational risks if the company is involved in "High-Impact Environmental Sectors" or has a questionable reputation. An Enhanced Due Diligence (EDD) report may be required to assess these risks.

Recommendations:

- FintechStream LLC should revise the proposal to include a comprehensive 3-year P&L forecast and address the financial viability concerns.

****Analysis of Proposal against Governance Context****

The proposal, PROJECT NEPTUNE, submitted by FintechStream LLC to the National Bank Group Business Development Group, is evaluated against the National Bank Group New Activity (NAC) Policy. The analysis highlights potential areas of concern and non-compliance with the governance context.

****Financial Viability (Rule 1.1)****

- * The proposal lacks a complete 3-year P&L forecast, which is a requirement for new partnership activities.
- * The estimated 10% IRR in Year 1 is below the minimum threshold of 12% without a written waiver from the Divisional CFO.

****Reputational Risk (Third-Party) (Rule 2.4)****

- * The proposal does not explicitly mention partnership with entities involved in "High-Impact Environmental Sectors." However, it is essential to confirm that FintechStream LLC is not involved in such sectors to ensure compliance.

****Data Sovereignty (Rule 3.2)****

- * The proposal states that client credit scores will be stored on FintechStream's proprietary servers in a Caribbean-based data center. This raises concerns about data sovereignty, as the data center may not be located within a Tier-1 jurisdiction.